



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510 Fax: +66-2721-3515

## SAFETY DATA SHEET

### Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

### 1. CHEMICAL IDENTIFICATION

#### Product Identification

Trade name: Coosonnice M

Product name: Mancozeb 80% WP

Product use: Fungicide

#### Relevant identified uses of the substance/mixture and uses advised against:

Commonly used in agriculture, professional turf management and horticulture.

### 2. HAZARD IDENTIFICATION

#### Adverse Human Health Effects:

Inhalation,

Inhalation of dust can cause the following: irritation of nose, throat, and lungs

Eye contact,

Direct contact with material can cause the following: moderate irritation

Skin contact,

Prolonged or repeated skin contact can cause the following: possible skin irritation-dermatitis due to skin sensitization

Ingestion,

Very low toxicity by ingestion

### 3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: Manganese ethylenebis (dithiocarbamate)(polymeric)complex with zinc salt

Common name: Mancozeb (proposed to ISO)

Chemical class: Dithiocarbamate-ETU

Empirical formula:  $[C_4H_6MnN_2S_4]_xZn_y$

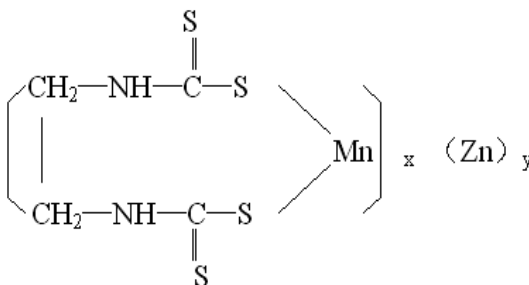
Molecular mass: 271.2



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Structural formula:



Ingredients and composition:

| Component                       | CAS No.   | % W/V  |
|---------------------------------|-----------|--------|
| Active ingredient<br>- Mancozeb | 8018-01-7 | 80 min |
| Inert ingredient                | -         | 20 max |

#### 4. FIRST AID MEASURES

##### Description of first aid measures

In case of eyes contact:

Flush with plenty of water at the least 15 minutes, call a physician.

In case of skin contact:

Wash affected skin areas thoroughly with soap and water. Consult a physician if irritation persists. Remove and wash contaminated clothing thoroughly. Do not take clothing home to be laundered.

In case of Inhalation:

Move subject to fresh air. Eye contact

Flush eyes with water. Consult a physician if irritation persists.

In case of Ingestion:

If swallowed, give 2 glasses of water to drink. See a physician. Never give anything by mouth to an unconscious person.

Note to Physician:

Have the person sip a glass of water if able to swallow. Do not give anything by mouth to an unconscious person.

Antidotes:

No specific antidotes are available against dithiocarbamate poisoning. If a large amount of mancozeb has been ingested in the last few hours, and if copious vomiting has not already occurred, the stomach must be emptied and steps taken to limit gastrointestinal absorption. If the patient is fully alert and nervous system depression is not anticipated, oral administration of Syrup of Ipecac is probably the best way to empty the stomach.



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## 5. FIRE FIGHTING MEASURES

|                            |                              |
|----------------------------|------------------------------|
| Flash point & Method:      | Not determined               |
| Flammable limits:          | 0.2 g/L (% by volume in air) |
| Auto Ignition Temperature: | Not determined               |

### FIRE FIGHTING HAZARDS& PROCEDURES:

#### General hazards:

Like most organic powers or crystals, under severe dusting conditions this material may form explosive mixtures in air.

#### Extinguishing media:

Use dry chemical carbon dioxide, water spray or foam.

#### Fire-fighting equipment/instruction:

Keep personnel removed and upwind of fire. Wear self-contained breathing apparatus. Wear full protective equipment. Use water spray. Runoff from fire control may be a pollution hazard if area is heavily exposed of fire and conditions permit. Let fire burn itself out since water may increase the contamination hazard.

#### Hazardous combustion products:

Hazardous gases/vapors produced in fire are H<sub>2</sub>S, CS<sub>2</sub>, and oxides of sulfur, nitrogen, and carbon.

## 6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

### Spill of Leak Procedures:

Remove source of heat, sparks, flame, impact, friction, or electricity. Dike spill. Prevent material from entering sewers, waterways, or low areas. Shovel or sweep up. Never return to container for reuse. Scoop into bags or boxes with plastic or aluminum shovel.

## 7. HANDLING AND STORAGE

### Precautions to be taken in handling and storage:

#### General information

Avoid breathing vapors or mist. Avoid breathing dust. Avoid contact with eyes, skin, or tobacco in areas where they may become contaminated with this material.

#### Handling:

Users should wash hands before eating, drinking, chewing gum, using tobacco or using toilet. Remove personal protective equipment immediately after handling this product. Wash the outside of gloves before removing. As soon as possible, wash thoroughly and change into clean clothing. Avoid dust regeneration.



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### Storage:

Store in a well ventilated place. Keep container tightly closed. Do not store or consume food, drink, or tobacco in areas where they may become contaminated with this material. Never allow the product to become wet during storage. This may lead to chemical changes with will reduce the effectiveness as fungicide and create flammable vapors.

## 8. EXPOSURE CONTROL/PERSONAL PROTECTION

### Personal Protective Equipment

#### Eyes Protection Requirements:

Wear protective eyewear to prevent contact with this substance.

#### Protective clothing Requirements:

Mixers and loaders must wear coveralls over long-sleeved shirt and long pant, shoe plus socks, protective eyewear, and chemical-resistant apron. Applicators and other handlers must wear coveralls over long-sleeved shirt and long pant, waterproof gloves, and shoe plus socks. Personnel re-entering treated areas that is permitted under the worker protection standard and that involves contact with anything that has been treated, such as plants, soil, or water coveralls over long-sleeved shirt and ling pants, waterproof gloves, and shoes plus socks. Any clothing or heavily contaminated with this product's concentrate must be discarded. Do not reuse them. Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables, use detergent and hot water. Keep and wash PPE separately from other laundry.

#### Respirator Requirements:

Wear dual cartridge respirator for dusts and mists.

#### Ventilation Requirements:

Provide local exhaust ventilation and/or general dilution ventilation to meet published exposure limits.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                          |                                        |
|--------------------------|----------------------------------------|
| Physical stage:          | Powered solid                          |
| Color:                   | Yellowish                              |
| Odor:                    | Sulfurous                              |
| Melting/ Freezing Point: | 192 °C (decomposed before this degree) |
| Moisture:                | ≤ 2%                                   |
| Bulk density:            | 0.35 - 0.5g/cm <sup>3</sup>            |



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## 10. STABILITY AND REACTIVITY

### Instability

This material is considered stable. However, keep away from moisture, heat or flame.

### Hazardous decomposition products

Thermal decomposition may yield the following: carbon disulfide-hydrogen sulfide

### Hazardous polymerization

Product will not undergo polymerization.

### Incompatibility

Avoid contact with the following: oxidizing agents-acids

## 11. TOXICOLOGICAL INFORMATION

Toxicity class WHO (a.i.): U

EPA formulation: IV

### Information on Acute toxicity for end product

Oral (rat): LD50 > 5,000 mg/kg

Dermal (rabbit): LD50 > 5,000 mg/kg

Inhalation (rat): LC50 (4h) > 5.14 mg/L

Skin irritation (rabbit): > practically non-irritating

Eye irritation (rabbit): > not irritating (EEC Classification)

Moderately irritating (US Classification)

### Carcinogenicity data

A two-year feeding study of mancozeb in rats produced an increased incidence of thyroid tumors at 750 ppm (29 mg/kg/day). No evidence of carcinogenicity was observed in long-term studies with mice.

### Mutagenicity data

Both mancozeb and ETU have been adequately tested in wide variety of in vitro and in vivo mutagenicity tests. The weight of the evidence of these tests indicates that mancozeb and ETU are not mutagenic in mammalian systems.

### Reproductive/Teratology data

No reproductive effects were seen below exposure levels high enough to produce non-reproductive toxic effects in two-generation reproduction studies of mancozeb or ETU in rats.

### Sensitization data

Mancozeb causes skin sensitization in guinea pigs when tested using the maximization procedure, but not when tested using the Buehler procedure. Consequently, mancozeb may have a weak potential for skin sensitization in humans.



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## 12. ECOLOGICAL INFORMATION

### Toxicity

Data referred to the mixture, Mancozeb:

#### **Fish – Acute toxicity:**

Rainbow trout,

LC50 (48 h): 1.9 ml/L

Bluegill sunfish,

LC50 (96 h): 1 ppm

#### **Birds – Acute toxicity:**

Japanese quail,

LD50 (10 days): >6400 mg/kg

Bobwhite quail,

Reproduction, NOAEL: 500 ppm

Mallard ducks,

LD50 (10 days): > 6400 mg/kg

Reproduction, NOAEL: 125 ppm

#### **Soil organism – Acute toxicity:**

Earthworm,

LC50 (14 days): > 299.1 mg/kg dry soil

Reproduction, NOEC: 20 mg/kg

#### **Arthropod – Acute toxicity:**

Honey bee,

LD50 (oral): > 209 µg /bee

LD50 (contact): > 400 µg /bee

## 13. DISPOSAL CONSIDERATIONS

### Waste Disposal Method:

Treatment, storage, transportation, and disposal must be in accordance with applicable federal, state/provincial, and local regulations. Do not flush to surface water or sanitary sewer system. Disposal of empty bag in a sanitary landfill or by incineration, or, if allowed by state and local authorities, by burned, stay out of smoke.



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#### 14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No: 3077

Dangerous Goods Class: 9

Packaging Group: III

Keep away from food, seed, and forage in storage and shipment. Avoid contact with skin and eyes.

#### 15. REGULATORY INFORMATION

Symbol: Xi

Indication of danger: Irritating substance

##### Label elements

##### **Risk phrases:**

R37: Irritating to the respiratory system.

R43: May cause sensitization by skin contact.

R50/53: Very toxic to aquatic organisms may cause long-term adverse effects in the aquatic environment.

##### **Safety phrases:**

S1/2: Keep locked up and out of reach of children.

S3/9/49: Keep only in the original container in a cool, well-ventilated place.

S13: Keep away from food, drink and animal feeding stuffs.

S20/21: When using do not eat, drink or smoke.

S23: Do not breathe sprays.

S24/25: Avoid contact with skin and eyes.

S29: Do not empty into drains.

S36/37/38: Wear suitable protective clothing, gloves and face protection.

S46: If swallowed seek medical advice immediately and show this container and label.

#### 16. OTHER INFORMATION

The data in this Material Safety Data Sheet relates only to the specific material designated herein and are based upon data believed to be correct.